Justin Mascari

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Education

University of California San Diego, San Diego, CA 2026
Bachelor of Science, Physics with Specialization in Astrophysics

Relevant Coursework:
Classical Mechanics, Electromagnetism, Fluids & Thermodynamics, Quantum Mechanics, Waves & Optics, Special Relativity, Circuits & Electronics, Mathematical & Computational Physics, Stars and Compact Objects, Black Holes, Observational Optical Research Lab, Intro to Python Programming, Statistical and Thermal Physics, Quantum Physics II, Modern Cosmology, Galaxies

California State University Northridge, Los Angeles, CA

Relevant Coursework:
Intro to Astrophysics

College of the Canyons, Santa Clarita, CA

Relevant Coursework:
Classical Mechanics, Electricity & Magnetism, Waves, Optics, Thermodynamics, Linear Algebra, Multivariable & Vector Calculus, and Differential Equations

Awards and Honors

Honorable Mention recipient of the Chambliss Astronomy Achievement Student Award 2026
UCSD Summer Program for Undergraduate Research in Science (SPURS) Award 2025
STARTastro Scholar 2024

Research

Team Nearby Galaxies Undergraduate Researcher Sept. 2024 - Present
University of California, San Diego, Department of Astronomy & Astrophysics
Mentors: Prof. Karin Sandstrom and Dr. Michael Busch

- Worked on mapping out CO-dark H_2 in the Triangulum Galaxy (M33) and the Andromeda Galaxy (M31)
  - Manipulated FITS data for dust and gas gathered from the Herschel Space Observatory (HSO), Very Large Array (VLA), and Atacama Large Millimeter Array (ALMA) for M33
  - Applied signal-to-noise cuts to carbon monoxide (CO) and dust data (3σ–10σ thresholds)
  - Worked on deriving a dust-to-gas ratio (DGR) and CO-H_2 conversion factor (X_{CO}) for M33 and M31
  - Performed radial profile analysis using a radial binning code to examine the DGR and $X_{CO,}$ as well as to inspect trends in metallicity.
  - Investigated the relationship between dust emission, atomic hydrogen (HI), and CO intensity to trace out the CO-dark H_2 abundance
  - Analyzed CO-dark gas fractions and explored how they are impacted by metallicity
  - Created a theoretical OH map for M33 to potentially draft a Green Bank Telescope (GBT) proposal that would help explore OH as a tracer for H_2

UCSD Gravitational Wave Paleontology Lab Oct. 2026 - Present
University of California, San Diego, Department of Astronomy & Astrophysics
Mentors: Prof. Floor Broekgaarden

- Contributing to GROWL, a collaborative catalog comparing binary population synthesis predictions across codes and physical assumptions to interpret gravitational-wave source populations.
- Modeled the evolution of a massive binary through stable mass transfer and common-envelope phases to binary black hole formation using COMPAS, and computed its gravitational-wave merger timescale; analyzed simulation output in Python (h5py, pandas, matplotlib).

Presentations and Posters

"Mapping the CO-dark H_2 Abundance in the Local Group"

Presentation:
2026 UCSD Undergraduate Research Conference, San Diego, CA Apr. 2026

Poster:
SPURS Poster Session, San Diego, CA Sept. 2025

Presentation:
2025 UCSD Summer Research Conference, San Diego, CA Aug. 2025

"Mapping CO-dark H_2 in Local Group Galaxies"

Poster:
247th American Astronomical Society Meeting, Phoenix, AZ Jan. 2026

"Tracing out "CO-dark" molecular hydrogen in M33"

Poster:
245th American Astronomical Society Meeting, National Harbor, MD Jan. 2025

"Creating a "CO-dark" H_2 Map of M33"

Presentation:
2024 UCSD Summer Undergraduate Research Conference, San Diego, CA Aug. 2024

Skills

Programming languages: Python, Mathematica, C++

Programming packages: astropy, numpy, pandas, matplotlib, scipy, reproject

Software: LaTeX, Overleaf, Github, SAOImageDS9, Jupyter Notebook

Computational and Analytical Skills:

- Calibration and data reduction of FITS images (bias subtraction, flat-fielding, SNR cuts)
- Multi-wavelength image regridding and coordinate reprojection
- 3D deprojection, centroiding, and radial profile binning
- Error propagation and statistical modeling of astrophysical data

Outreach, Leadership, & Experience

Contract Tutor for Unlimited Learning LLC Aug. 2026 - Present

- Work as a paid tutor for Unlimited Learning. Provide one-on-one math and physics tutoring for students at a high school and college level. Help students develop their confidence and foundational knowledge in their coursework.

STARTastro Tutor Nov. 2025 - June 2026

- Worked as a paid tutor for the STARTastro transfer program within the UCSD Astronomy and Astrophysics Department. Provided tutoring and mentorship to undergraduates at the lower and upper-division level. Offered homework help, exam preparation, and research advice.

MESA Math & Physics Tutor Aug. 2023 - Jan. 2025

- Worked as a paid tutor for the Mathematics, Engineering, Science Achievement (MESA) at College of the Canyons.
- Tutored and mentored STEM students at a community college level by reviewing concepts and problems related to math and physics.
- Hosted workshops and study groups where I offered students supplementary instruction to help better their understanding of material taught in their math and physics courses.

Barrio Logan Science & Art EXPO Apr. 2025

- Volunteered to help the UCSD Astronomy & Astrophysics Department in setting up and presenting demonstrations related to astronomy